# S3 — Equilibrium and Acid-Base Chemistry

## 3.1 Dynamic Equilibrium

**Dynamic equilibrium:** A state reached in a closed system when the rate of the forward reaction equals the rate of the reverse reaction, so the concentrations of reactants and products remain constant but are not necessarily equal. Both forward and reverse reactions continue to occur at the molecular level.

Conditions for dynamic equilibrium:

- The system must be closed (no substances can enter or leave).
- The forward and reverse rates must be equal.
- The concentrations of all species remain constant over time.
- The reaction must be reversible.

### Le Chatelier's Principle

**Le Chatelier's principle:** If a system at equilibrium is subjected to a change in conditions (concentration, pressure, or temperature), the position of equilibrium will shift in the direction that tends to oppose the change, so as to partially restore the original conditions.

| Change | Effect on equilibrium position | Effect on K |
|---|---|---|
| Increase concentration of a reactant | Shifts to the right (towards products) | No change |
| Decrease concentration of a product | Shifts to the right | No change |

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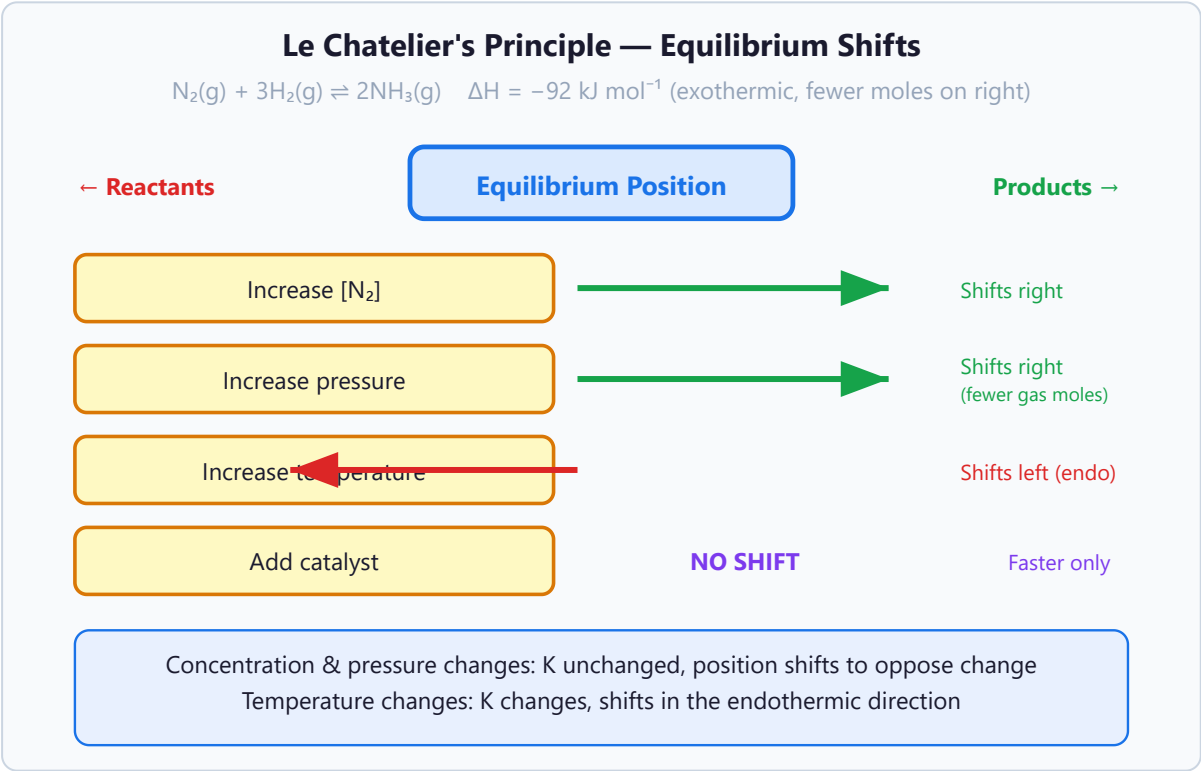
| | | |
|---|---|---|
| Increase pressure | Shifts towards the side with fewer moles of gas | No change |
| Increase temperature | Shifts in the endothermic direction | Changes (increases if forward reaction is endothermic) |
| Add a catalyst | No change (speeds up both forward and reverse equally) | No change |

**Exam Tip:** A catalyst does not change the position of equilibrium or the value of K. It only increases the rate at which equilibrium is reached. Temperature is the only factor that changes the value of the equilibrium constant.




# 3.2 Equilibrium Constants

## The Equilibrium Constant $K_c$

For a general reaction: $aA + bB \rightleftharpoons cC + dD$

$$K_c = [C]^c[D]^d / [A]^a[B]^b$$

where square brackets denote equilibrium concentrations in mol $dm^{-3}$. $K_c$ is constant at a given temperature. A large $K_c$ means the equilibrium lies to the right (products favoured); a small $K_c$ means it lies to the left (reactants favoured).

Only species in the aqueous or gaseous phase are included in the expression. Pure solids and pure liquids are excluded because their concentrations are effectively constant.

## The Equilibrium Constant $K_p$

For gaseous equilibria, $K_p$ is expressed in terms of partial pressures:

$$K_p = p_C{}^c \cdot p_D{}^d / (p_A{}^a \cdot p_B{}^b)$$

**Partial pressure:** The pressure that a gas would exert if it alone occupied the total volume of the mixture at the same temperature. The partial pressure of gas A is: $p_A = x_A \times P_{total}$, where $x_A$ is the mole fraction of A.

**Mole fraction:** $x_A$ = moles of A / total moles of all gases.

The total pressure equals the sum of all partial pressures (Dalton's law): $P_{total} = p_A + p_B + p_C + ...$

**Exam Tip:** When calculating $K_p$, pay careful attention to units. The units of $K_p$ depend on the stoichiometry of the reaction. Work out the overall power of pressure in the numerator minus the denominator to determine the unit (e.g. Pa, kPa, atm, or dimensionless).

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**Equilibrium Constants — What K Tells Us**

**K « 1**
Reactants favoured

**K ≈ 1**
Neither side favoured

**K » 1**
Products favoured

**$K^c$ (concentrations)**

aA + bB ⇌ cC + dD

$K^c = [C]^c[D]^d / [A]^A[B]^b$

**$K^p$ (partial pressures)**

For gaseous equilibria

$K^p = p^{cc}\, p^{dd} / p^{AA}\, p^{bb}$

**What affects the value of K?**

✓ Temperature changes K (shifts equilibrium in endothermic direction)

✗ Concentration, pressure, and catalysts do NOT change K

# 3.3 Acids and Bases

## Brønsted-Lowry Theory

**Brønsted-Lowry acid:** A proton ($H^+$) donor.

**Brønsted-Lowry base:** A proton ($H^+$) acceptor.

**Conjugate acid-base pair:** Two species that differ by a single proton. When an acid donates a proton, it forms its conjugate base. When a base accepts a proton, it forms its conjugate acid.

Example: $HCl + H_2O \rightarrow H_3O^+ + Cl^-$. Here HCl is the acid (donates $H^+$) and $H_2O$ is the base (accepts $H^+$). $Cl^-$ is the conjugate base of HCl, and $H_3O^+$ is the conjugate acid of $H_2O$.

## Strong and Weak Acids

**Strong acid:** An acid that fully dissociates (ionises) in aqueous solution. Examples: HCl, $H_2SO_4$, $HNO_3$. For a strong monobasic acid, $[H^+]$ = [acid].

**Weak acid:** An acid that only partially dissociates in aqueous solution. An equilibrium exists between the undissociated acid and its ions. Examples: $CH_3COOH$, $H_2CO_3$.

## pH Calculations

$$pH = -\log_{10}[H^+]$$

$$[H^+] = 10^{-pH}$$

**Strong acid pH:** For a strong monobasic acid at concentration c, $[H^+] = c$, so $pH = -\log c$.

**Strong base pH:** For a strong base like NaOH at concentration c, $[OH^-] = c$. Use $K_w$ to find $[H^+]$: $[H^+] = K_w / [OH^-]$, then calculate pH.

## The Acid Dissociation Constant $K_a$

For a weak acid $HA \rightleftharpoons H^+ + A^-$:

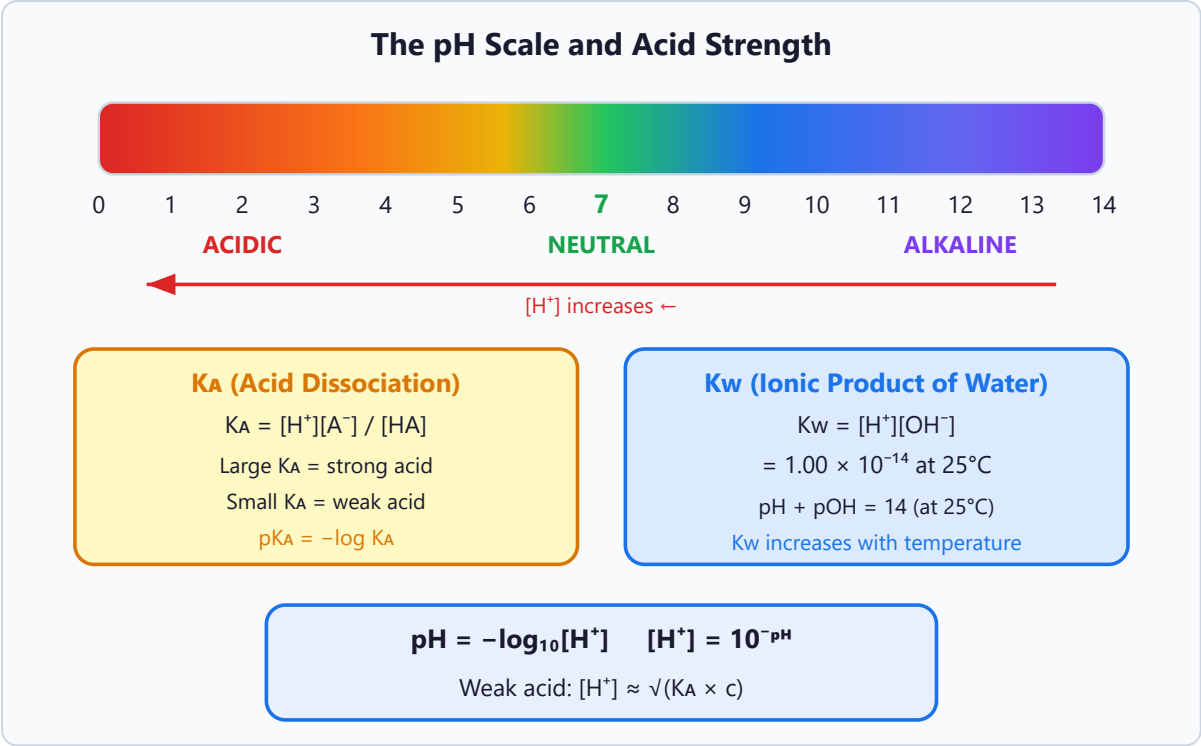
$$K_a = [H^+][A^-] / [HA]$$

For a weak acid, we assume $[H^+] = [A^-]$ (because the autoionisation of water contributes negligibly) and that the extent of dissociation is so small that the equilibrium concentration of HA is approximately equal to the initial concentration. Therefore:

$$[H^+] \approx \sqrt{(K_a \times c)}$$

$$pK_a = -\log_{10} K_a$$

The smaller the $K_a$ (or larger the $pK_a$), the weaker the acid.




## The Ionic Product of Water $K_w$

**$K_w$:** The ionic product of water. At 25 °C, $K_w = [H^+][OH^-] = 1.00 \times 10^{-14}$ mol$^2$ dm$^{-6}$. $K_w$ increases with temperature because the dissociation of water is endothermic.

In pure water at 25 °C: $[H^+] = [OH^-] = 1.00 \times 10^{-7}$ mol dm$^{-3}$, giving pH = 7.00. At higher temperatures, $K_w$ is larger, so $[H^+]$ is greater and the pH of pure water is less than 7 — but the water is still neutral because $[H^+] = [OH^-]$.

# 3.4 Buffer Solutions

**Buffer solution:** A solution that resists changes in pH when small amounts of acid or base are added. It consists of a weak acid and its conjugate base (or a weak base and its conjugate acid) in roughly equal concentrations.

## How Buffers Work

An acidic buffer (e.g. ethanoic acid + sodium ethanoate) contains a large reservoir of both HA molecules and $A^-$ ions:

- **When acid is added:** The extra $H^+$ ions react with the conjugate base $A^-$ to form HA. This removes the added $H^+$, so the pH changes very little: $H^+ + A^- \rightarrow HA$
- **When base is added:** The extra $OH^-$ ions react with the weak acid HA to form $A^-$ and water. This removes the added $OH^-$: $OH^- + HA \rightarrow A^- + H_2O$

The buffer works because HA and $A^-$ are present in large, roughly equal concentrations, so small additions of $H^+$ or $OH^-$ cause only tiny changes to the ratio $[HA]/[A^-]$ and hence tiny changes in pH.

## Calculating Buffer pH

Starting from $K_a = [H^+][A^-] / [HA]$, rearranging gives:

$$[H^+] = K_a \times [HA] / [A^-]$$

Then $pH = -\log[H^+]$. The Henderson-Hasselbalch equation provides an equivalent form:

$$pH = pK_a + \log_{10}([A^-] / [HA])$$

When $[HA] = [A^-]$, $pH = pK_a$. This is also the point at which the buffer has maximum buffering capacity. When choosing a buffer for a particular pH, select a weak acid whose $pK_a$ is close to the desired pH.

**Exam Tip:** In buffer calculations, you can use moles instead of concentrations for [HA] and $[A^-]$ because they are in the same solution volume, so the volume cancels in the ratio. Be careful when the buffer is made by adding a strong base to an excess of weak acid — work out the moles of HA remaining and $A^-$ formed.

**Worked Example — pH of a Weak Acid:**

Calculate the pH of 0.100 mol $dm^{-3}$ ethanoic acid ($K_a = 1.74 \times 10^{-5}$ mol $dm^{-3}$).

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Assumptions: $[H^+] = [A^-]$; $[HA]_{eq} \approx [HA]_{initial}$

$[H^+] = \sqrt{(K_a \times [HA])} = \sqrt{(1.74 \times 10^{-5} \times 0.100)} = \sqrt{(1.74 \times 10^{-6})} = 1.32 \times 10^{-3}$ mol dm$^{-3}$

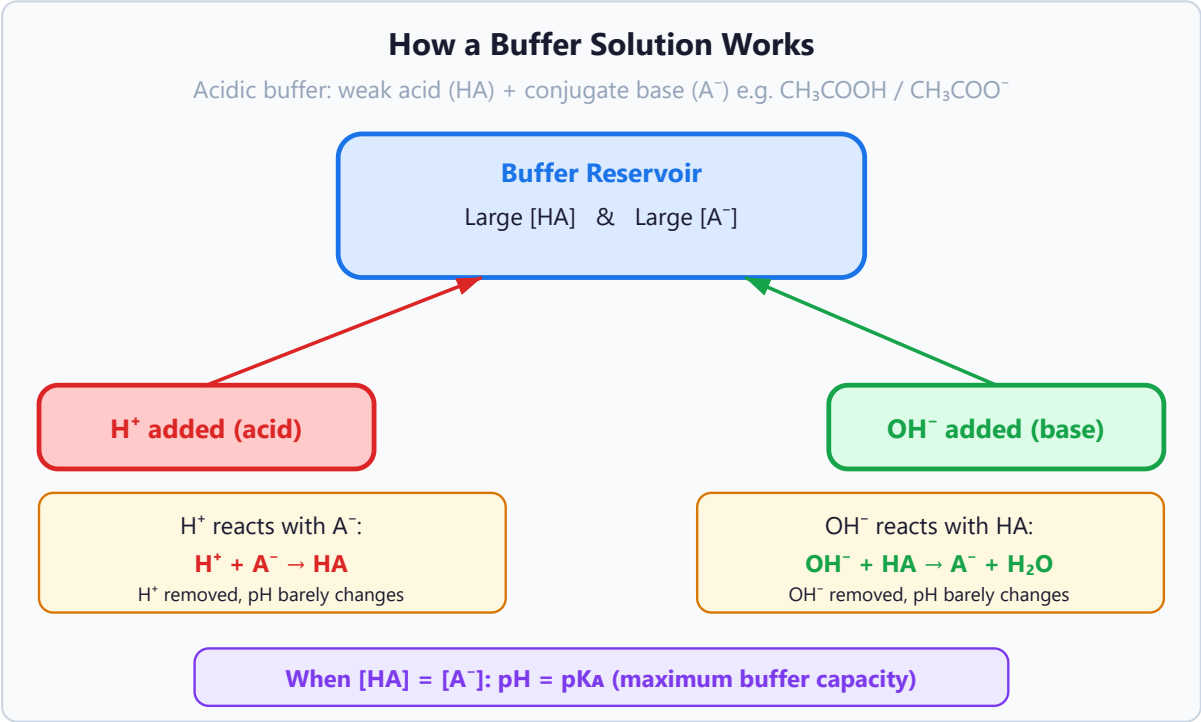
pH = $-\log(1.32 \times 10^{-3})$ = **2.88**

**Worked Example — Buffer pH:**

A buffer is made from 0.200 mol $CH_3COOH$ and 0.100 mol $CH_3COONa$. $K_a = 1.74 \times 10^{-5}$.

$[H^+] = K_a \times [HA] / [A^-] = 1.74 \times 10^{-5} \times 0.200 / 0.100 = 3.48 \times 10^{-5}$ mol dm$^{-3}$

pH = $-\log(3.48 \times 10^{-5})$ = **4.46**




# 3.5 Titration Curves and Indicators

## Titration Curves

A pH titration curve plots pH against volume of reagent added. The shape depends on the strengths of the acid and base:

| Combination | pH at equivalence point | Suitable indicator |
|---|---|---|
| Strong acid + strong base | 7 | Any indicator with a range between pH 3–11 (e.g. phenolphthalein, methyl orange) |
| Strong acid + weak base | < 7 (acidic) | Methyl orange (range pH 3.1–4.4) |
| Weak acid + strong base | > 7 (basic) | Phenolphthalein (range pH 8.3–10.0) |
| Weak acid + weak base | ≈ 7 (variable) | No suitable indicator (gradual pH change, no sharp equivalence point) |

Key features of a strong acid–strong base curve: the pH changes slowly initially, then very sharply near the equivalence point (a near-vertical section), then levels off. The equivalence point is at the centre of the vertical section.

For a weak acid–strong base titration: the initial pH is higher (because the acid is weak), the curve rises more gradually, and the equivalence point pH is above 7 because the salt formed is slightly basic (the conjugate base of the weak acid hydrolyses).

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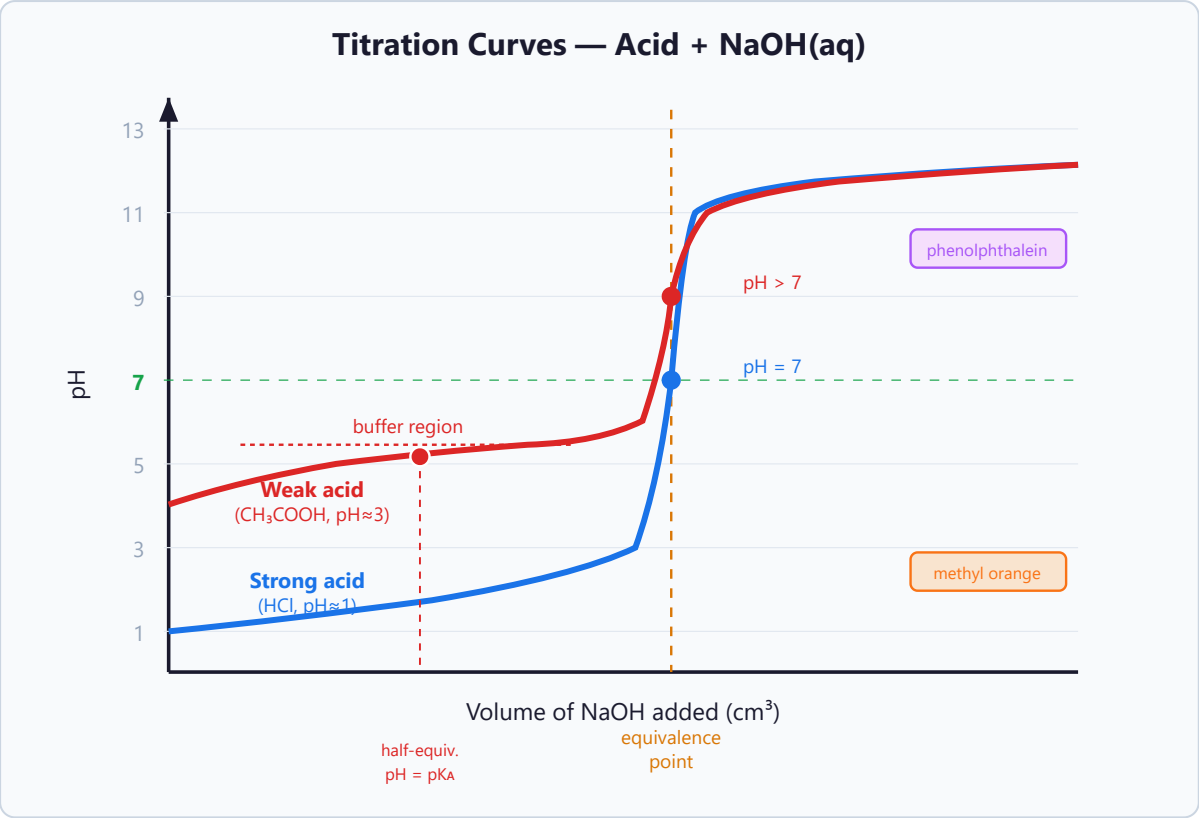


## The Half-Equivalence Point

At the half-equivalence point (half the volume needed to reach the equivalence point), exactly half the weak acid has been neutralised, so [HA] = [$A^-$]. At this point, pH = $pK_a$. This provides an experimental method for determining $K_a$.

## Indicators

**Indicator:** A weak acid (HIn) whose conjugate base ($In^-$) has a different colour. The indicator changes colour over a range of about 2 pH units centred on its $pK_{In}$ value. A suitable indicator must have its colour change range within the steep (vertical) portion of the titration curve.

The indicator equilibrium: HIn(aq) ⇌ $H^+$(aq) + $In^-$(aq). In acidic solution [HIn] dominates (colour A); in basic solution [$In^-$] dominates (colour B). The colour change occurs at pH ≈ $pK_{In}$.

## Neutralisation

Neutralisation is the reaction between an acid and a base to form a salt and water. For strong acid + strong base: $H^+$(aq) + $OH^-$(aq) → $H_2O$(l). The standard enthalpy of neutralisation for this reaction is

approximately $-57.1$ kJ $mol^{-1}$. For reactions involving weak acids or bases, the enthalpy of neutralisation is less exothermic because some energy is used to complete the dissociation of the weak acid or base.

**Exam Tip:** When sketching titration curves, label the axes clearly (pH on y-axis, volume of titrant on x-axis). Mark the equivalence point, the initial pH, and the buffer region (the relatively flat section before the equivalence point in a weak acid–strong base titration). The buffer region is where pH changes least per unit volume of titrant added.

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